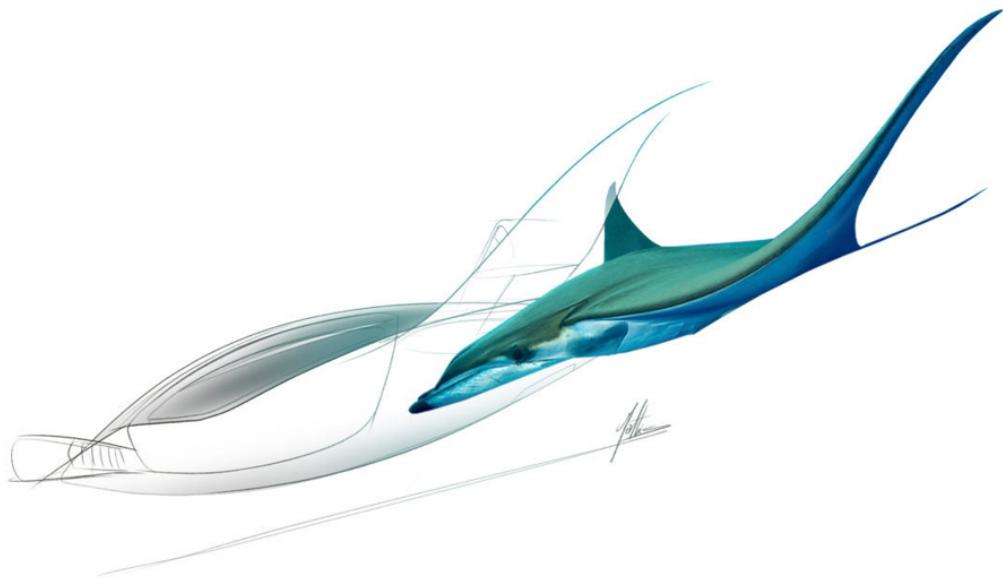


Global Innovators' Challenge

*How can biomimicry influence aircraft wing design
to achieve carbon neutrality in aviation?*



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1. Introduction

According to a roadmap produced by Sustainable Aviation (Fig. 1), the most effective contribution to IATA’s industry-wide Fly Net Zero 2050 target is the use of Sustainable Aviation Fuel (SAF) [1]. However, I wish to explore the next most impactful solution, “fleet upgrades with future aircraft types”. In this paper, I will explore various biomimetic approaches to optimising wing shape to boost lift and efficiency. I will then propose a new design, the Shark-Skin Blended Wing Body, justifying its strengths and limitations.

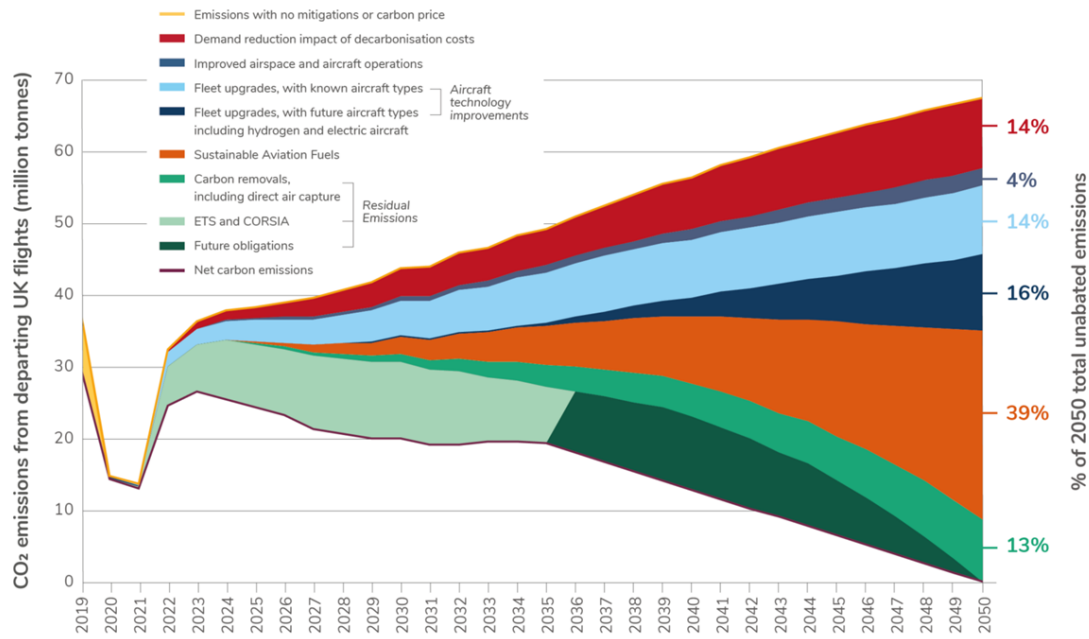


Figure 1 – Sustainable Aviation Net Zero Carbon Road-Map

2. Blended Wing Bodies (BWBs)

BWB concepts involve a continuous and wide wing that appears moulded into the fuselage. As will be seen, these designs are effective at boosting lift as they increase the aspect ratio (wingspan divided by mean chord length) of the wing, so a greater mass of air is diverted per unit time. Many BWBs are inspired by the manta ray, who use wide lifting surfaces with large planform areas to cruise efficiently in-flight.

2.1. Inspired by the Snow Goose

Fleets of snow geese are often seen flying in a large V-shape. The purpose is to enable following geese to surf on the wakes of the lead bird, minimising fatigue as the birds gain ‘free lift’ during long flights. Inspired, Delft University have developed the Flying-V concept (Fig. 2). This design consumes 20% less Jet-A fuel than the A350, because the cabin, cargo hold and fuel are all located in the wing [2]. This consequently decreases

structural weight and maximises lift. High-lift devices (flaps and slats) are not needed due to its high lift-to-drag (L/D) ratio, further reducing weight and maintenance costs.

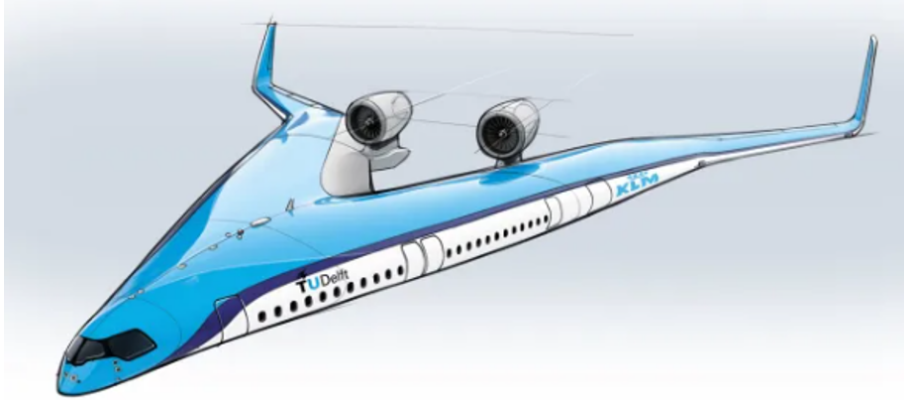


Figure 2 – Delft University’s Flying-V design

2.2. Inspired by the Eagle and Falcon

Another existing idea is Airbus’ “Bird of Prey” (Fig. 3). Its blended wing-to-fuselage joint mimics the aerodynamic arch of an eagle or falcon. Each feather can be individually controlled mid-flight to best control the balance of lift generated and drag minimised. An article by the IMechE reports that the split tail may facilitate finer control of this balance, while the absence of a vertical stabiliser further mitigates induced and parasite drag [3]. However, since its unveiling in 2019, little further research has been conducted by Airbus, so its benefits remain to be verified.



Figure 3 – Airbus’ “Bird of Prey” concept

2.3. Limitations

Despite the aforementioned benefits of BWBs of improving L/D ratios and hence fuel efficiency, some designs may be incompatible with existing airport infrastructure. The

main issue here is the spacings of airport gates. The keynote speaker at Imperial College’s 2024 Aeronautics Research Showcase, Sue Partridge, estimated that a typical gate width is approximately 36 m. Therefore, folding wings, akin to those installed on B777Xs, are required. This idea will be explored below. Radical redesigns like the Flying-V are unlikely to be certified to carry passengers till at least the 2040s.

3. Morphing Wings

These can be defined as wings which change in shape during flight, perhaps through use of control surfaces (flaps, ailerons etc.), but also encompass designs that intentionally bend or flex, like birds in nature.

3.1. Inspired by the Albatross

The albatross is a seabird that can “lock” its wings at the shoulder to fly long distances with minimal fatigue, and “unlock” these to navigate wind speeds better. Such flexible wingtips are analogous to car suspension, dampening the effects of wake turbulence while reducing drag. Wingtips are not uncommon today: the B777X’s wingtips add 7 m to its 65 m wingspan, offering 10% lower fuel consumption while still adhering to Class E airport dimension requirements [4]. Returning to Sue Partridge: she heads the Airbus “Wing of Tomorrow” (WoT) programme, which aims to create a leaner, longer, and lighter wing inspired by nature, with a high aspect ratio. In particular, the use of foldable wingtips means the wingspan can be minimised during gate time and taxi, and fully extended for maximum surface area (and lift creation) in flight.

Airbus is no stranger to copying the albatross: a more direct example is their AlbatrossONE demonstrator (Fig. 4). This involves a similar aero-elastic hinged wingtip to the WoT. This has several benefits, including reducing wing loading and decreasing the drag by-products of vortices, to ultimately cut CO₂ emissions by 20% compared to current medium-range airliners [5].



Figure 4 – Airbus’ AlbatrossONE project

3.2. Limitations

While the utility of semi-aeroelastic wingtips have been confirmed through wind-tunnel testing by Airbus on its eXtra Performance Wing, these tests did not account for the many flexible joints, actuators, and composite skins needed to safely integrate the wingtips. This adds weight, partly offsetting the CO₂ savings initially suggested above. Furthermore, adoption will be slow due to extensive certification requirements, though perhaps less so than BWBs due to the scope of innovation of the latter.

4. Inspired by the Shark

At the 2000 Olympics, 15 swimming world records were broken, more than any previous game. What 13 of the records had in common was that their swimmers wore the newly-introduced Speedo Fastskin swimsuits [6]. These were composed of dermal denticles (Fig. 5), akin to the skin of sharks. These create a low-pressure zone (“leading-edge vortex”) as water moves over the skin, enabling them to be thrust-enhancing, propelling the swimmer forward. It also disrupted skin friction drag.

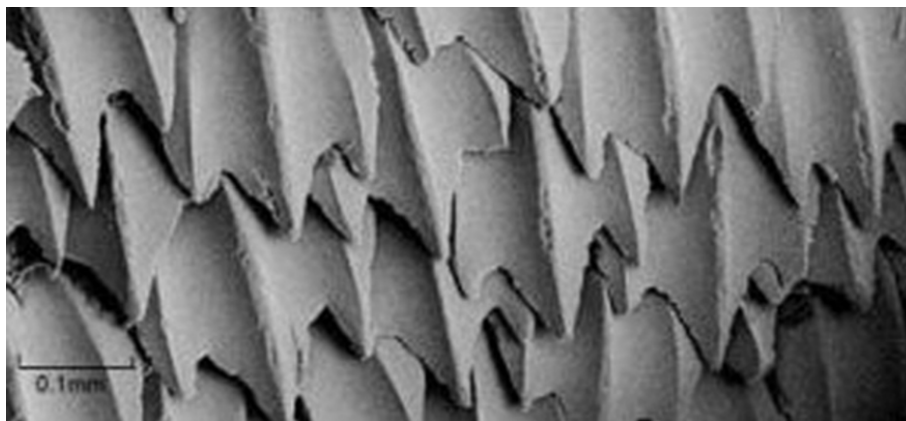


Figure 5 – Speedo Fastskin swimsuit denticles

This technology has been developed to create riblets – streamwise surface striations which are aligned with the local velocity of air, reducing friction drag from a turbulent boundary layer. It has been found that using equilateral triangle riblets across an aircraft’s fuselage and wings can reduce friction drag by 8% [7].

Lufthansa have picked up on this through their AeroSHARK skin which will coat their entire cargo fleet, with estimated annual savings of <11,700 tons of CO₂ emissions [8].

5. Proposal

Having evaluated the above advancements in wing design, I now propose the **Shark-Skin Blended Wing Body** (Fig. 6). This consists of a BWB shape similar to a manta ray. The whole fuselage and wing will be covered with riblet microtextures, just like the Speedo swimsuits. It will have a high aspect ratio, achieved via a long wingspan. To fit in airport gates, foldable wingtips will be installed as found on the WoT. The individually-controlled feathers from the Bird of Prey will not be featured, as this will require excessive additional testing for its worth. Like the Bird of Prey, it could lack a vertical stabiliser, though as a side-effect stability and control will be compromised.

Shark-Skin Blended Wing Body

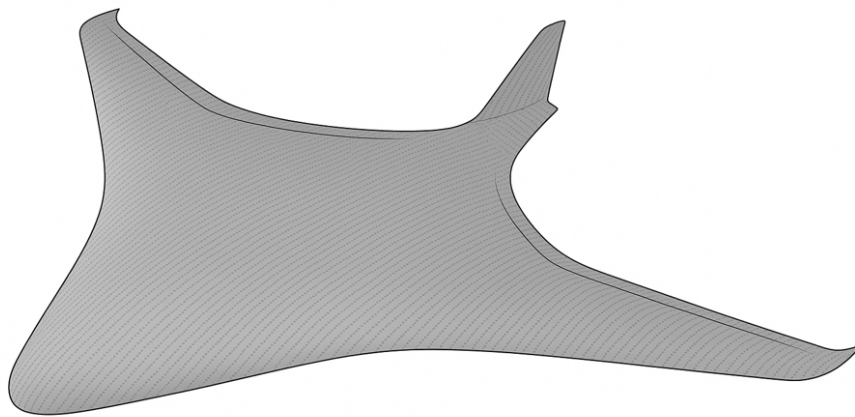


Figure 6 – Mock-up draft of the Shark-Skin Blended Wing Body

Many of these design considerations (riblets, high aspect ratio, wingtips etc.) are highly effective at lowering the friction drag experienced by the aircraft, while simultaneously boosting lift to achieve greater L/D ratios, saving fuel consumption for a given distance. With all the above features combined, CO₂ could fall by as much as 40–50%. However, this is a crude estimation taken using the figures listed earlier in the paper. Many issues persist, such as stability concerns, budget, weight, and whether such a design could be certified in time for the 2050 carbon neutrality goal. Yet, I think these are necessary trade-offs, most of which are applicable to any inventive design. Either way, the whole aviation industry must do something before it's too late, and while biomimicry isn't and shouldn't be the only solution to sustainability, we ought to take more note of how nature approaches the numerous challenges it faces.

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